



# COMSATS University Islamabad

## Department of Computer Science

### Course Description Form (CDF)

#### Course Information

Course Code: **CSC336**

Course Title: **Web Technologies**

Credit Hours: **3(2,1)**

Lecture Hours/Week: **2**

Lab Hours/Week: **3**

Pre-Requisites: **CSC241–Object Oriented Programming**

#### Course Objectives

- To explain the fundamental concepts of web architectures and its applications;
- To provide an understanding of planning, designing and publishing a multi-page website;
- To provide hands-on experience of client-side technologies such as HTML, JavaScript, CSS, and React.js
- To provide practical experience in developing dynamic web applications using modern full-stack technologies

#### Course Contents

This course introduces the modern web technologies used for web development. Topics include Overview of Web Platforms; Web Architectures; Markup Languages; Styling; Client-Side Languages; Server-Side Technologies; Use of Databases in Web-based Applications; REST APIs; Application Security, Domain setup and Deployment.

#### Unit wise Major Topics:

| Unit | Topic                                                                                                                                                                                                                                                                                                                                       | No. of Teaching Hours |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| 1.   | Web Development: Overview, Platforms, Terminologies, Standards & Constraints, Applications Categories, and Development Environments. Tiered Web Architecture: One-Tier, Two-Tier, and N-Tier.                                                                                                                                               | 2                     |
| 2.   | Markup Languages: HTML and HTML5, Tags for headings, links, images, paragraphs and lists. Block Level vs. Inline Elements, Semantic HTML, HTML Forms                                                                                                                                                                                        | 2                     |
| 3.   | Styling: Cascading Style Sheets (CSS), Box Model, Positioning, CSS3, Layout Design using Flex Box & Grid Box, Responsive Design, Framework Fundamentals: Bootstrap or Tailwind Framework                                                                                                                                                    | 6                     |
| 4.   | Modern JavaScript (ES6+) for Web Development: Core JavaScript concepts (variables, data types, operators, scope), Functions, Closures, Important built-in methods (map, filter, reduce, forEach). DOM Manipulation, Event Handling, Regular Expressions for validation and parsing, JSON, Promise, and AJAX using Fetch API.                | 5                     |
| 5.   | Server Side Programming: Request Handling, HTTP Status Codes and Common Headers, Sessions, Cookies, Middleware, Database Connectivity with PostgreSQL or MongoDB, ORM, Response Generation using Templates, REST-API Development, Session and Token Based Authentication. Implementation using NodeJS/Express or Python/Django or FAST API. | 9                     |
| 6.   | Front-end Development using React.js: Introduction, Rendering Approaches, Components, Event Handling, Hooks, State Management, Routing, API integration                                                                                                                                                                                     | 4                     |

|                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                |                                                                                               |                                |                 |                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------------------------------|-----------------|----------------------------------|
| 7.                                                                                                                                                                                                                                                                                                                                                                                                                | Application deployment and applying application level security |                                                                                               |                                | 2               |                                  |
| Total Contact Hours                                                                                                                                                                                                                                                                                                                                                                                               |                                                                |                                                                                               |                                | 30              |                                  |
| Mapping of CLOs and PLOs                                                                                                                                                                                                                                                                                                                                                                                          |                                                                |                                                                                               |                                |                 |                                  |
| Sr.#                                                                                                                                                                                                                                                                                                                                                                                                              | Unit #                                                         | Course Learning Outcomes                                                                      | Blooms Taxonomy Learning Level | PLO             |                                  |
| CLO's for Theory                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                |                                                                                               |                                |                 |                                  |
| CLO-1                                                                                                                                                                                                                                                                                                                                                                                                             | 1-7                                                            | Explain fundamental concepts, principles, architectures, and technologies of web development. | Understanding                  | 2               |                                  |
| CLO-2                                                                                                                                                                                                                                                                                                                                                                                                             | 2-5                                                            | Apply web development principles in designing responsive and interactive web interfaces.      | Applying                       | 4               |                                  |
| CLO-3                                                                                                                                                                                                                                                                                                                                                                                                             | 4-6                                                            | Analyze approaches for developing functional client-side and server-side web applications.    | Analyzing                      | 3               |                                  |
| CLO's for Lab                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                |                                                                                               |                                |                 |                                  |
| CLO-4                                                                                                                                                                                                                                                                                                                                                                                                             | 2-6                                                            | Implement functional web applications using appropriate front-end and back-end technologies.  | Applying                       | 4               |                                  |
| CLO-5                                                                                                                                                                                                                                                                                                                                                                                                             | 1-7                                                            | Develop functional web applications using appropriate development technologies.               | Creating                       | 5               |                                  |
| CLO Assessment Mechanism                                                                                                                                                                                                                                                                                                                                                                                          |                                                                |                                                                                               |                                |                 |                                  |
| Assessment Tools                                                                                                                                                                                                                                                                                                                                                                                                  | CLO-1                                                          | CLO-2                                                                                         | CLO-3                          | CLO-4           | CLO-5                            |
| Quizzes                                                                                                                                                                                                                                                                                                                                                                                                           | Quiz 1                                                         | Quiz 2                                                                                        | Quiz 3 & 4                     | -               | -                                |
| Assignments                                                                                                                                                                                                                                                                                                                                                                                                       | -                                                              | Assignment 1&2                                                                                | Assignment 3&4                 | Lab Assignments | -                                |
| Midterm Exam                                                                                                                                                                                                                                                                                                                                                                                                      | Mid Term Exam                                                  | Mid Term Exam                                                                                 | -                              | Mid Term Exam   | -                                |
| Final Term Exam                                                                                                                                                                                                                                                                                                                                                                                                   | Final Term Exam                                                |                                                                                               |                                | -               | Lab Project/ Lab Final Term Exam |
| Text and Reference Books                                                                                                                                                                                                                                                                                                                                                                                          |                                                                |                                                                                               |                                |                 |                                  |
| Textbooks:<br>1. Web Design Playground: HTML & CSS the Interactive Way, Paul McFedries, Manning, 2019.<br>2. Eloquent JavaScript by Marijn Haverbeke (4th edition, 2024)<br>3. Learning React: Modern Patterns for Developing React Apps 2nd Edition by Alex Banks, Eve Porcello, 2020.<br>4. Learn React Hooks by Daniel Bugl, 2019.                                                                             |                                                                |                                                                                               |                                |                 |                                  |
| Reference Books:<br>1. Web Engineering, Sahil Rai, Kuk University Notes, 2020.<br>2. Web Programming with HTML5, CSS, and JavaScript, John Dean, Jones & Bartlett Learning 2018.<br>3. Murach's Modern JavaScript by Mary Delamater (2024)<br>4. Mastering FastAPI: Build High-Performance APIs with Python — Aimed at building scalable, high-performance APIs with advanced features.<br>5. FastAPI (O'Reilly). |                                                                |                                                                                               |                                |                 |                                  |

